IWATANI CORPORATION · CONFIDENTIAL WORKING DRAFT

Supply Chain Slide: Source Citations

Verified citation for every claim on the slide, plus a slide-ready footer and three corrections

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry, leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. Two claims on the slide are negative findings that cannot be cited and should be attributed to NRI analysis.

18 — Source Citations for the Supply Chain Map Slide

Verified citations for every claim on the "Manufacturing process & supply chain map" slide, plus three factual corrections that should be made before it is shown.

Three corrections first

1. The bottom callout says "Iwatani's **Japan-origin** precision slitting technology" — this is not supportable

Iwatani's own Precision Stainless Steel page names its precision slitting **processing sites** as exactly two, both in **China**:

> "**Suzhou Iwatani Metal Products Co., Ltd.** and **Zhongshan Iwatani Co., Ltd.** are Iwatani group companies responsible for precision slitting of precision stainless steel and non-ferrous metal materials." — Iwatani — Precision Stainless Steel **[P]**

No Japan-based slitting site is named on either the Precision Stainless Steel page or the Stainless Steel page. The company says it "operates a precision slitting business and has established a system for supplying customers around the world" — but the disclosed physical slitting assets are Suzhou and Zhongshan, plus a **precision pressing and insert moulding plant in Thailand** added in 2025 (FY2026 Investors' Guide **[P]**).

Why this matters acutely: you are presenting to Iwatani. They know where their own slitting lines are. Describing the capability as "Japan-origin" when their published processing sites are in Jiangsu and Guangdong is the kind of detail that undermines everything else on the slide.

Suggested rewording: "Iwatani's existing precision slitting capability (Suzhou and Zhongshan, China; precision pressing in Thailand)" — or simply "Iwatani's existing precision slitting capability," avoiding the geography claim entirely.

2. The callout frames coil slitting as "the entry point" — but India already has it

Both Indian precision-strip mills already slit **and** explicitly name bellows as a served application:

- **IUP Jindal Metals & Alloys** — three Brodeur slitting lines with shimless tooling, edge-rounding machine, slits to 3.5 mm width; application list includes **"Flexi metal tubes / bellows"** (brochure **[P]**)
- **Quality Foils (India)** — slit edge from 5.0 mm; product listing includes **"Metallic Bellows"** and **"Bellows for precision measuring instruments"** (products **[P]**, Kompass listing **[C]**)

The middle box on the slide is accurate — no bellows manufacturer slits in-house. But the inference that slitting capacity is therefore the gap does not follow, because independent Indian mills supply it. **The actual gap is one stage upstream: grade and tolerance class.** No Indian mill publishes AM350, and the only published Indian thickness tolerance at 0.10 mm gauge is ±0.020 mm against Alleima's ±0.001 mm.

Suggested rewording of the callout: "Indian bellows makers buy slit strip rather than slitting in-house — but the harder gap is upstream: no Indian mill produces AM350 or matches global tolerance class. That grade-and-tolerance gap, not slitting capacity, is Iwatani's entry point."

3. "Diaphragm leaf stamping of seamless tube" conflates the two manufacturing routes

These are alternative routes, not one process:

Route	Process	Feedstock
Edge-welded (semiconductor/ UHV)	Blank/stamp **diaphragm leaves** from foil, stack, weld ID then OD	Thin foil
Formed / convoluted (industrial)	Roll strip to **tube**, longitudinal seam weld, then hydroform or mandrel-form convolutions	Slit strip

Sources: MW Components — edge welding technology **[P]**; Triad Bellows — how metal bellows are made **[P]**.

Suggested rewording: "Diaphragm leaf stamping (edge-welded route) or tube forming and hydroforming (convoluted route)."

Minor, optional: "Bellows manufacturer" as a stage label is still an actor rather than a process — "Final Assembly" would keep the chevron sequence logically consistent, since the box content already reads "Final assembly into a sellable bellows unit."

Slide-ready source footer

One line, sized for a slide footer:

> **Sources:** Company websites and technical brochures (Jindal Stainless, IUP Jindal Metals & Alloys, Quality Foils India, Well Tech Metal Bellows, Fluidyne Engineers, Bhastrik Mechanical Labs, KSM, Technetics, Witzenmann, Ulbrich, TOKKIN, Alleima, Lamineries Matthey, SMC, VAT Group); MCA/ROC filings; Indian customs data (HS 7219/7220); NRI analysis.

If you have room for two lines, add:

> **Note:** Grade and tolerance data are from each mill's own published specifications. Indian bellows manufacturer capability claims are from company listings and have not been independently audited.

That second line matters — several of the Indian capability claims (Well Tech's 0.05–0.2 mm AM350, Bhastrik's 0.127 mm) come from IndiaMART/TradeIndia listings rather than audited sources, and saying so protects you if a claim is later challenged.

Backup slide: per-stage citations

Drop this on a backup/appendix slide, or keep it in the notes pane.

Stage	Claim on slide	Verified source	Grade
1. Raw Material Procurement	Melting, rolling, coiling of stainless / nickel alloy / titanium into master coil	Jindal Stainless product brochure — EAF/AOD/VOD melt, hot Steckel + tandem mills, 20-Hi Sendzimir cold rolling	**[P]**
	Materials used: stainless, nickel alloy, titanium	Valqua — bellows — full material list: SUS 304/304L/316/316L/321/347, AM-350, Inconel, Hastelloy, titanium, Monel	**[P]**
	"Procured from other domestic companies and from imports as well (e.g. AM350)"	**Domestic:** IUP Jindal brochure (0.03–1.5 mm, austenitic only); Quality Foils products (0.10–4.00 mm). **Imports:** no Indian mill publishes AM350 — see Ulbrich AM350, UPM AM350, TOKKIN 350, Lamineries Matthey AM350	**[P]**
	AM350 is the semiconductor bellows reference grade	VAT Group 04.2 transfer valve — specifies bellows material **AISI 633 (AM 350)**	**[P]**
2. Precision Slitting	Cutting master coil into narrow, tight-tolerance precision-width strips	Hempel Special Metals — precision slit strip — service centre explicitly serving metallic bellows and expansion joint manufacturers, from 0.05 mm	**[P]**
	Tolerance benchmark	Alleima precision strip brochure — to 0.015 mm at **±0.001 mm**	**[P]**
	India tolerance comparator	Quality Foils products — published **±0.020 mm at 0.10 mm** gauge	**[P]**
	"No domestic bellow manufacturer has in-house precision slitting"	Negative finding across all Indian bellows makers surveyed (Well Tech, Fluidyne, Bhastrik, Metallic Bellows, MB Metallic Bellows, Flexpert) — none publish or claim in-house coil slitting. **Absence of evidence from the survey conducted, not a citation to a source stating this**	**finding**
	Slit strip is available domestically	IUP Jindal — slitting; IUP Jindal facilities	**[P]/[U]**
3. Forming and Stamping	Diaphragm leaf stamping (edge-welded route)	MW Components — edge welding technology — hydraulic stamping of diaphragms	**[P]**
	Tube forming / hydroforming (convoluted route)	Triad Bellows; Comflex	**[P]/[C]**
	In-house at Indian plants	Flexpert — metal expansion joints (2–20 ply laminated); Witzenmann India brochure (HYDRA hydraulic forming)	**[P]**
4. Joining	TIG / laser welding of leaf pairs	MW Components — plasma, laser, arc, EB welding ID-to-ID then OD-to-OD	**[P]**
	Micro-plasma variant, India	Bhastrik — edge welded bellows listing — micro-plasma welding with anti-stress fixtures, diaphragm 0.127–0.30 mm	**[U]**
	In-house at Indian plants, 0.05 mm capability	Well Tech — mechanical seal bellows (0.05–0.2 mm, AM350); Fluidyne — diaphragms (0.05 mm and 0.10 mm)	**[U]/[P]**
5. Finishing and testing	Cleaning, passivation, leak testing, dimensional QC, certification	KSM — cleaning & product cleanliness — ISO Class 6 welding/assembly, ISO Class 5 final pack; Witzenmann high-purity brochure — UHV/UHP cleaning, optional RGA certificate	**[P]**
	Leak-rate qualification bar	VAT 04.2 valve — $<1\times10^{-9}$ mbar·l/s	**[P]**
	"Limited cleanroom & advanced testing capacity" in India	Negative finding: no Indian ISO Class 5/6 bellows cleanroom or RGA capability identified in this survey. Closest adjacent capabilities: APT leak-tests chambers to 1×10^{-12} Torr·l/s; KASFAB Tools has cleanrooms + precision welding. Fluidyne publishes in-house helium leak, spring rate, squirm and burst testing (about)	**finding + [P]/[C]**
6. Final assembly	Final assembly into a sellable bellows unit	Technetics — BELFAB — contract assembly incl. design, fabrication, welding, upper-level assembly, testing, packaging	**[P]**
	End fittings and hardware range	Witzenmann — edge welded bellows; Metallic Bellows India — gimbal, hinged, pressure-balanced, jacketed configurations	**[P]/[U]**

Stage	Claim on slide	Verified source	Grade
7. End-customers	Semiconductor	Hudson Technologies (slit valve, wafer lift, load lock, chamber lift); KSM SEMICON West 2026 (etch, CVD, implant, CMP, litho)	**[P]**
	Aerospace & Defence	Technetics (reservoirs, toggle switches, cold plates, fuel drains, engine kiss seals); Fluidyne certifications (HAL Nasik — LCA Mk1A, HTT-40; SIATI award for bellows)	**[P]**
	Nuclear & Power	Fluidyne certifications (BARC, NPCIL registrations); MB Metallic Bellows (thermal & nuclear power)	**[P]**
	Oil & Gas	Technetics (actuators, connectors, couplings, feedthroughs, gas lines); Flexpert (petrochemical, oil & gas distribution)	**[P]**
	Heavy Industrial (cement, steel, marine, sugar)	Flexpert (steel, cement, shipbuilding); MB Metallic Bellows (minerals & metals, hydrocarbons)	**[P]**
	Automotive	Autocar Professional — Witzenmann India — 10m+ exhaust decouplers produced for India	**[C]**
	Chemical	Flexpert (chemical processing); Well Tech (ANFD, chemical industry seals)	**[P]/[U]**
Aftermarket (not on slide — worth adding)	Bellows are a designed-in consumable, re-ordered for the life of the tool	SMC XGT — 3M cycle life, user-replaceable bellows; Irie Koken — 10,000 to 10,000,000 cycles	**[P]**

How to handle the two "negative findings"

Two boxes on the slide state an absence rather than a fact:

- "No domestic bellow manufacturer has in-house precision slitting"
- "Limited cleanroom & advanced testing capacity"

Neither can be cited to a source, because no document exists stating what a country lacks. They are conclusions from the scope of research conducted. **The honest way to present them on a slide is to attribute them to your own analysis rather than implying a citation:**

> Marked ● = NRI analysis based on published capability of surveyed Indian manufacturers; not independently audited.

Add that marker to those two boxes specifically. It is a small thing that materially changes how the slide survives challenge — a client who asks "who says?" gets a straight answer rather than a scramble.

Grade key for the appendix

Grade	Meaning
[P]	Company's own website, brochure, or filing — directly citable
[C]	Credible trade or business press
[U]	Trade directory listing (IndiaMART, TradeIndia) — indicative, not audited
finding	Conclusion from this study's own research scope, not a citation

Full source register: `06-source-register.md` (by topic) · `17-value-chain-bibliography.md` (by value chain stage).